

Business Brief

Company: WellLife General Hospital

Reporting Period: 2021-2024

Analysis Type: Descriptive Analytics

Prepared for: Management

Prepared by: Data Analytics Team

1. Background

WellLife General Hospital has collected data spanning from 2021 to 2024 to assist with hospital planning and resource allocation. The dataset contains key patient information such as Patient_ID, Age, Diagnosis, Gender, and Admission_Date. Additionally, diagnoses are grouped into two categories: Patient Cases and Chronic Cases. Management requires a comprehensive overview of patient distribution, diagnosis trends, and demographic factors to improve resource allocation for the upcoming years.

2. Problem Statement

Management currently lacks a consolidated, summarized view of patient admissions, diagnosis trends, and demographic data for 2021-2024. Specifically:

- There is no clear understanding of patient distribution by age, gender, or diagnosis group.
- Trends in patient admissions over time are not readily visible.
- Diagnosis types (chronic vs. patient cases) need further clarification.
Without this information, management is unable to effectively plan for the allocation of hospital resources or assess trends across different patient groups.

3. Analysis Objectives

The objective of this analysis is to describe and summarize the distribution of patients from 2021 to 2024. This analysis will:

- Present the total number of patients recorded from 2021 to 2024.
- Show the distribution of patients by age and gender.
- Summarize patient distribution by diagnosis group (Chronic Cases, Patient Cases).
- Highlight trends in patient admissions by year.

4. Key Business Questions (WHAT-focused)

Using the dataset, the following questions should be addressed:

- **Total Patients**
 - What is the total number of patients admitted from 2021 to 2024?
- **Age Distribution**
 - What is the average age of patients admitted during the period
 - How are patients distributed across various age?
- **Gender**
 - What is the gender distribution of patients?
- **Diagnosis Group**
 - How are patients distributed across the diagnosis groups
- **Admission Trends**
 - How do patient admissions vary across different years (2021-2024)?
 - What is the year-on-year growth rate of patient admissions?
 - Are there any notable seasonal or annual trends in admissions?
 - How many patients do we expect to be admitted during the first three months of 2025?

5. Expected Outcome

At the end of this analysis, management should be able to:

- Understand the distribution of patients by age, gender, and diagnosis group.
- Identify patient admission trends for better resource planning.
- Make informed decisions about hospital resource allocation based on patient demographics and diagnosis categories.

6. Note to Management

This analysis will provide a clear understanding of what the data shows about patient admissions over the past few years, which will help in planning and optimizing hospital resources for the future.

